

4.4 The management of tectonic hazards

The **primary** and **secondary responses** to a tectonic event are also known as immediate and long-term responses. The primary responses are those that occur in the days or weeks after a disaster happens. They mainly involve the search and rescue process and helping those who have suffered because of the disaster. The secondary responses go on for months and even years after a disaster. They involve the rebuilding of buildings and other infrastructure, while also rebuilding the economy.

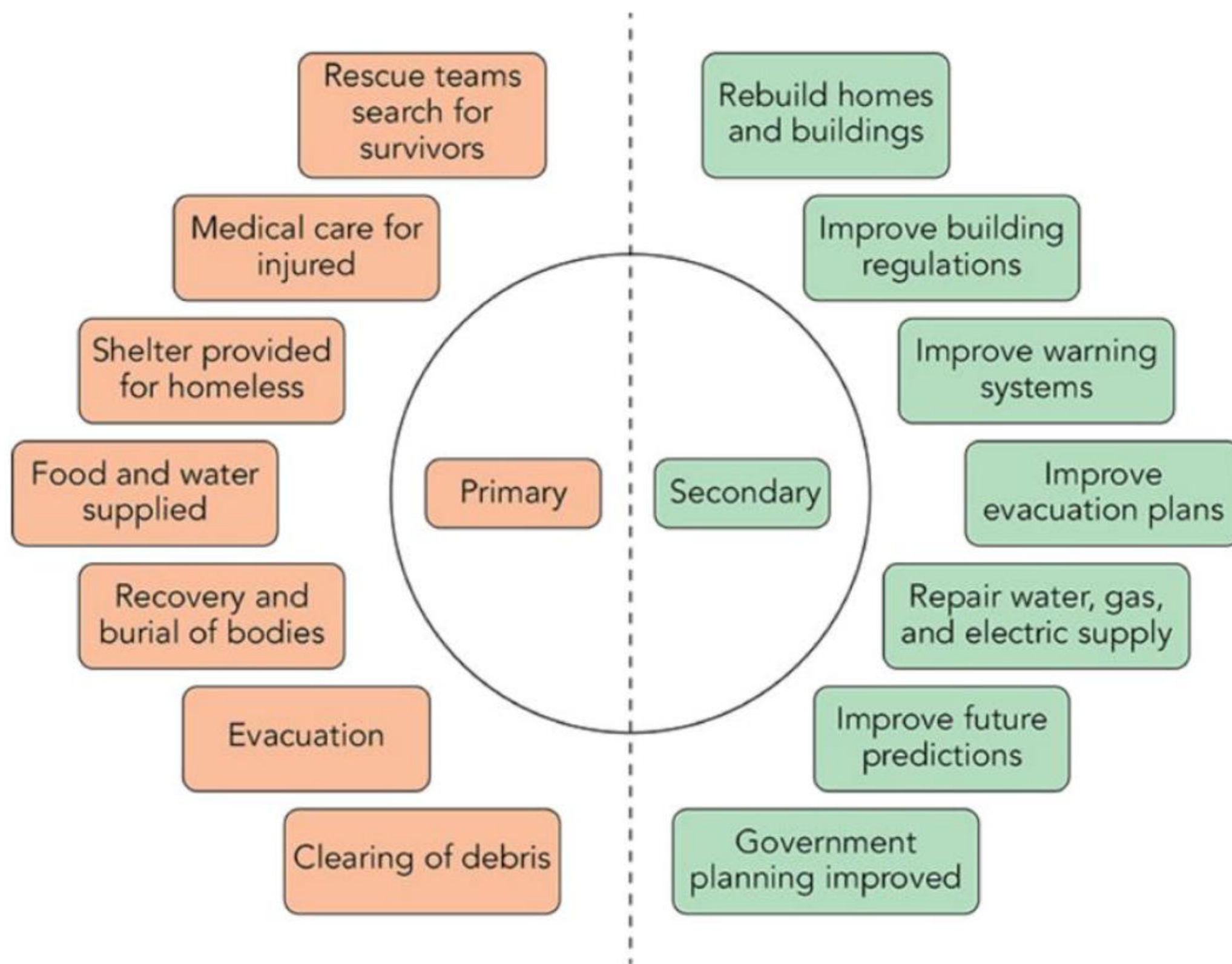


Figure 4.3: Primary and secondary responses to tectonic hazards

Despite growing knowledge of earthquakes and volcanoes, including the use of technology such as **seismographs** and **hazard mapping**, it is not yet possible to predict exactly when an earthquake or volcanic eruption will occur. There are a number of strategies and techniques we can use to manage the impact of tectonic hazards, including prediction, protection, and planning (often shortened to 'PPP').

Monitoring and prediction using technology

The monitoring and prediction of tectonic hazards is dependent on a few factors. These include the availability of technology such as seismographs, tilt meters, gas detectors, **Geographical Information Systems (GIS)**, and satellite imagery to track changes in tectonic plates. Changes in the Earth's crust indicate potential volcanic eruptions or an increased possibility of earthquakes.

Predictions can also rely on knowledge of the tectonic history of an area. Where a region regularly suffers from earthquakes, or a volcano is known to be active, local knowledge and observations (for example, observing increased steam emissions or cracks in a slope) can be used to monitor and predict events. These are, however, less reliable than using technology to determine when an event may occur.

Predictions are not yet fully precise and can result in false alarms, which risks people taking them less seriously and ignoring future warnings. The information gathered using technology is still valuable when mapping potential hazards, however, and can be used to indicate when areas are becoming unsafe.

REFLECTION

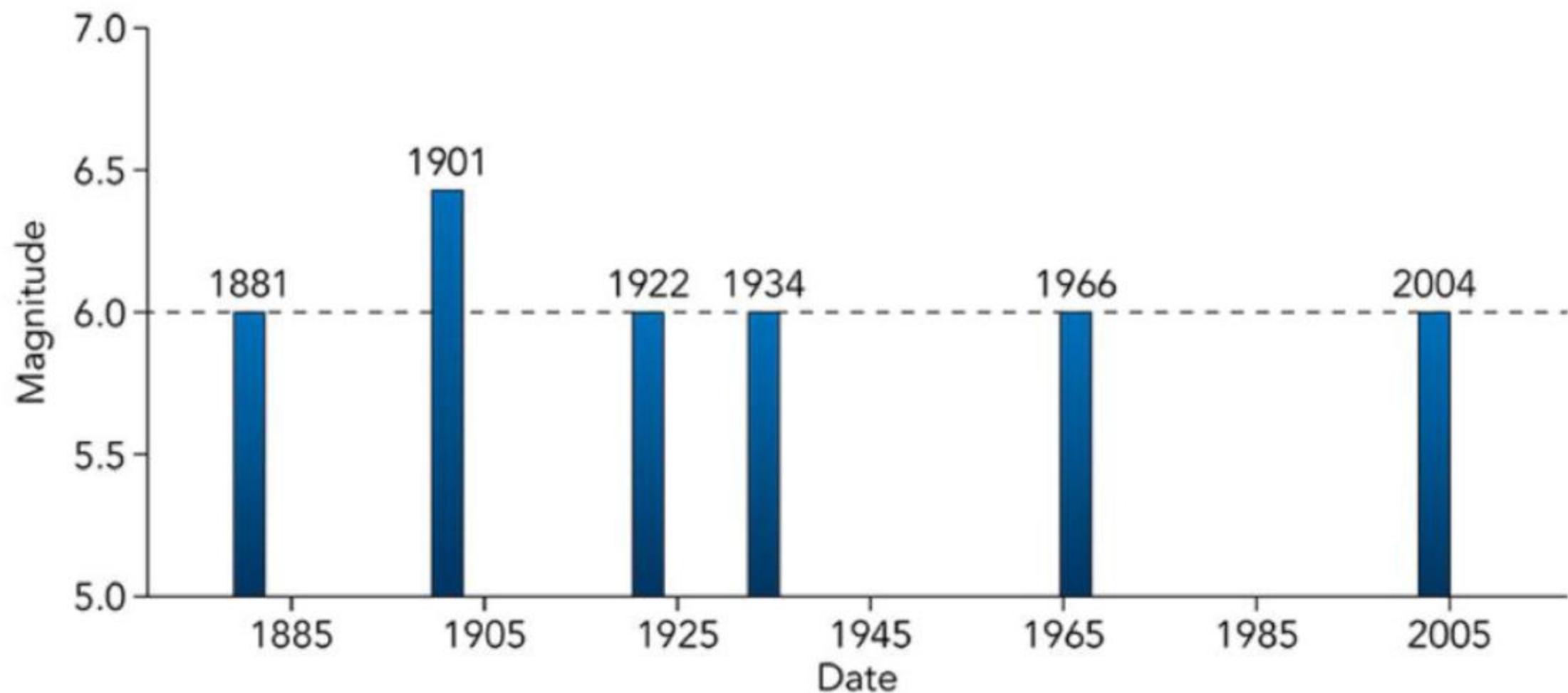
When completing a research task, do you find it difficult to find relevant information? Do you think that if you tried different search terms you would find more information? Did you discuss your findings with a study partner to improve your results? Which strategy helped you most?

- a Explain how the evacuation of an earthquake zone can help to reduce the impacts.
- b In pairs, discuss how prediction, planning, and prevention (PPP) could decrease the need for the identified responses to tectonic hazards. Write a 200 to 250-word summary to show what you think the impact of a good PPP system would be on the required responses to a tectonic event.
- c In pairs, discuss why PPP is important in the management of tectonic hazards. Consider the role that each 'P' plays in the protection of the population and infrastructure from tectonic hazards.

TASK 2

In pairs or small groups:

- a Research different prediction strategies to identify a tectonic risk.
- b Using your research, create a poster showing three prediction strategies that you have identified. Your poster should explain how they can be used to reduce the impacts of a tectonic hazard.



Source A: The last six earthquakes in Parkfield, California, USA

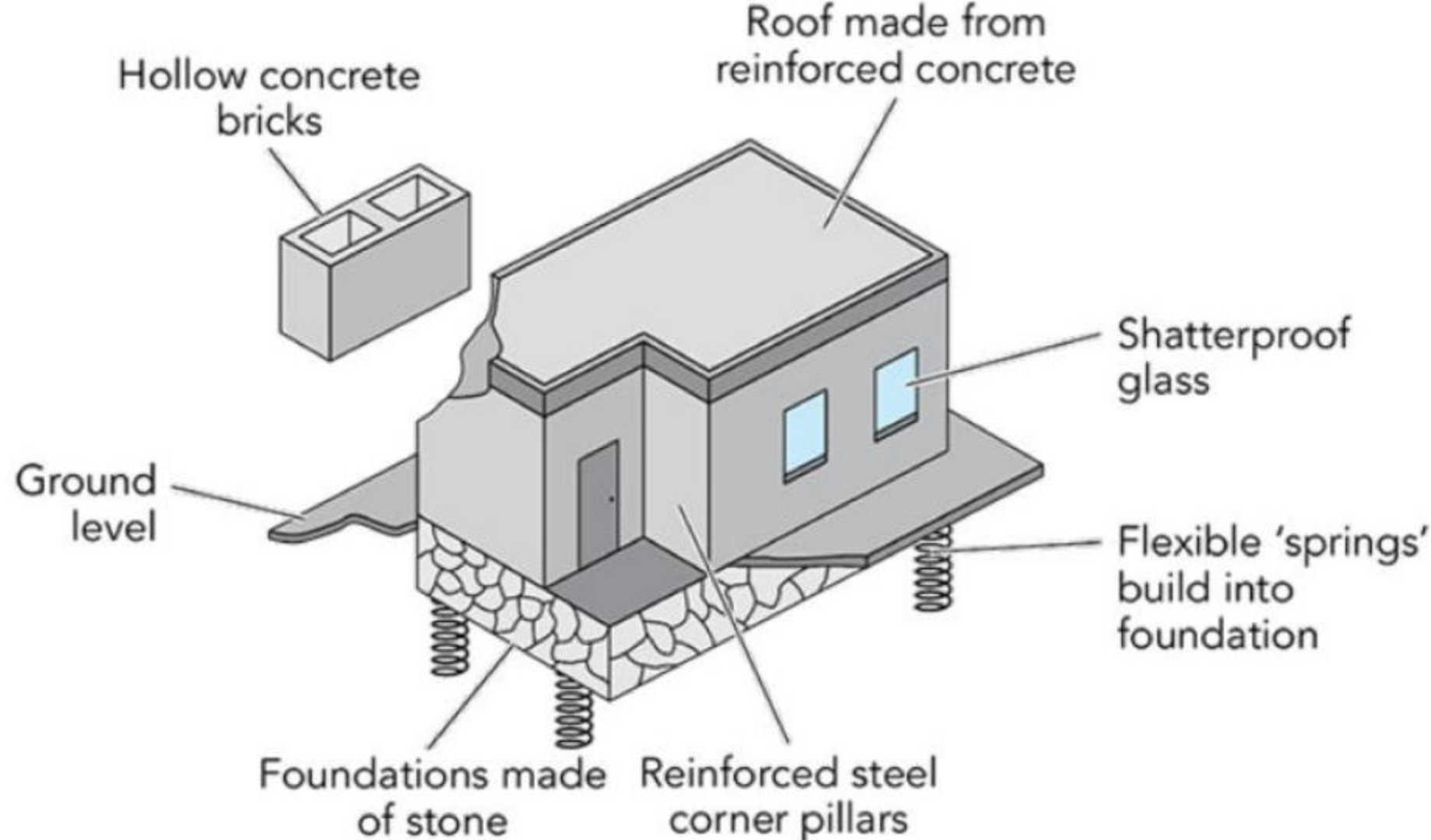
TASK 3: Study Source A

- a **Identify** which year had the earthquake with the highest recorded magnitude for Parkfield.
- b **Calculate the time between:**
 - i The first and second earthquake
 - ii The third and fourth earthquake
 - iii The fifth and sixth earthquake.
- c Refer to your answers to b to explain why the frequency of earthquakes cannot be used as reliable data to predict the next earthquake.
- d With reference to data from Source A, is it possible to predict the magnitude of the next earthquake? Explain your answer.

Protection

In addition to early warning systems, areas at risk from a tectonic hazard can take other steps to protect the population. These include:

- a Building structures to withstand the shaking caused by earthquakes, or the weight that ash can add to a roof. This helps to prevent the collapse of buildings.
- b Introducing land-use zoning regulations, which includes risk assessments that result in hazard maps. These maps are used to ensure that building does not take place in high-risk zones. Instead, infrastructure is built in areas that are considered safer.



Source **B**: Preparing buildings for earthquakes

156

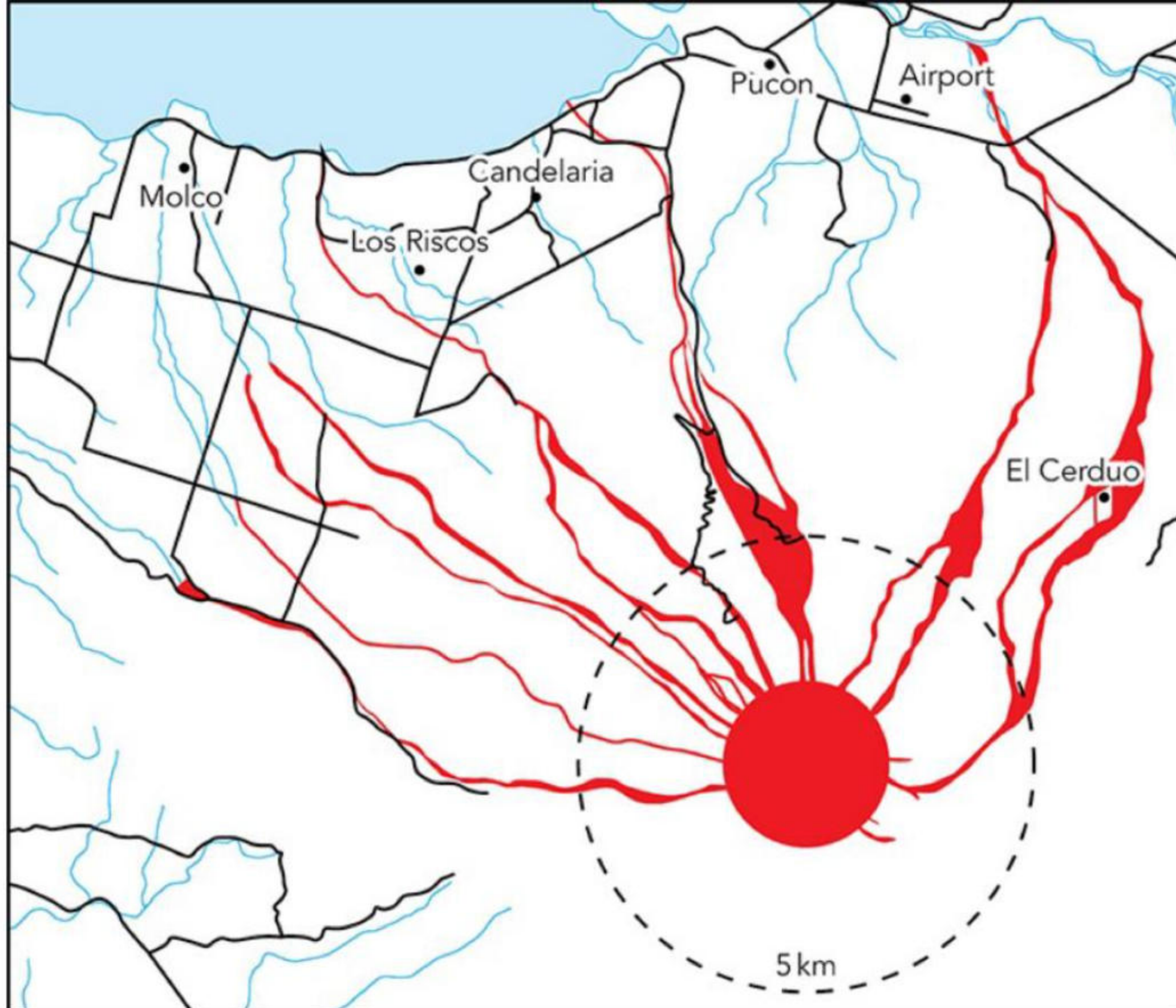
TASK 4: Study Source **B**

- a** Suggest how flexible 'springs' in the foundation will help to protect the building during an earthquake.
- b** This building is only one storey high. Suggest how that helps to reduce the risks of injury and death during an earthquake.
- c** If the earthquake triggered a tsunami, what is the potential risk that this building poses to inhabitants?

Planning

Good planning can reduce the impacts of a tectonic event. For an area to be well prepared, clear emergency plans with rescue teams and equipment, recovery plans, storage of food, water, and shelter for use in the event of an earthquake or volcanic eruption are all required to assist with saving people and reducing the recovery time.

Insurance and aid can help areas recover. Insurance shares the costs of recovery between people who are impacted by the event and those who are not. This helps with supplying money to rebuild and recover. In areas where insurance is not available, governments may ask for international aid (help given from high-income countries to countries that need assistance) and donations to allow for rescue and long-term recovery.



Source C: A GIS hazard map of volcano Villarrica in Chile, South America, in 2015. The map shows the areas highest risk from lahars and lava flow.

TASK 5: Study Source C

- Give the meaning of the term lahar.
- Describe the areas at highest risk from the eruption of volcano Villarrica.
- The main airport is in the town of Pucon, to the north of the volcano. Explain how this may increase the risk of managing the evacuation of the population during an eruption by Villarrica.
- Work in pairs and write down three pieces of advice that you would give a person to help them survive a volcanic eruption. Then share your three ideas with the class and create a list with all the ideas for the class.

PROJECT

In pairs or small groups, carry out research into earthquake-resistant buildings. You will use this information to design an earthquake-resistant building. Consider the following factors:

- the movement of the Earth during an earthquake, including risk of hazards like liquefaction
- the strength and flexibility of materials used to construct a building
- the weight of material used and the harm it can cause during failure
- the height of the building
- the most stable shape for an earthquake-resistant building.

Draw an annotated diagram of your concept building, clearly explaining the different factors that you have included to reduce the risk of building failure during an earthquake. As an additional challenge, you could construct a model of the building concept that you have come up with to show how it is earthquake resistant.